

# CHAO REN

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👤 *Director, Department of Information and Communication Engineering, College of Electronics and Information Engineering (CEIE), Sichuan University, Chengdu, China.*

👤 *Associate Research Professor, CEIE, Sichuan University, Chengdu, China.*

## 🎓 EDUCATION BACKGROUND

**Sichuan University**, Sichuan, Chengdu Sep. 2012 – Jun. 2017  
Communications and Information Systems *Ph.D*

**University of California, San Diego (UCSD)**, USA Sep. 2015 – Sep. 2016  
Electrical and Computer Engineering *Joint Ph.D.*

**Sichuan University**, Sichuan, Chengdu Sep. 2008 – Jun. 2012  
Electronic Information Engineering *Bachelor*

## 👥 WORK EXPERIENCE

**Sichuan University**, Sichuan, Chengdu Apr. 2025 – Present  
College of Electronics and Information Engineering *Director*

**Sichuan University**, Sichuan, Chengdu Sep. 2021 – Present  
College of Electronics and Information Engineering *Associate Research Professor*

**Sichuan University**, Sichuan, Chengdu May 2018 – Aug. 2021  
College of Electronics and Information Engineering *Assistant Research Professor (Full-time Research Track)*

**Sichuan University**, Sichuan, Chengdu Jul. 2017 – Apr. 2018  
College of Electronics and Information Engineering *Assistant Researcher*

## 💬 ACADEMIC SERVICE

### 1. Academic Journal Service:

- **Guest Editor:** Electronics (ISSN: 2079-9292), Applied Sciences–Basel (ISSN: 2076-3417), Sensors (ISSN: 1424-8220), AI (ISSN: 2673-2688), etc.
- **Youth Editorial Board Member:** Journal of Engineering Science (ISSN: 2075-4914), Journal of Data Acquisition and Processing (ISSN: 1004-9037), Journal of Naval Aviation University (ISSN: 2097-1427).
- **Executive Editorial Board Member:** Communication Technology (ISSN: 1002-0802).

### 2. Academic Conference Service:

- International Conference on Image, Vision and Computing (ICIVC 2025), **Publication Chair and Conference Moderator**
- International Conference on Machine Vision, Image Processing and Imaging Technology (MVIPIT 2025), **Publicity Chair**
- International Conference on Robotics, Artificial Intelligence and Intelligent Control (RAIIC 2025), **Publicity Chair**
- IEEE International Conference on Intelligent Perception and Pattern Recognition (IPPR 2025), **Technical Committee Chair**
- International Conference on Image, Signal and Computing (ICISC 2023), **Program Committee Member**

- International Conference on Optics and Image Processing (ICOIP 2023), **Publication Chair**
- International Conference on Machine Vision, Image Processing and Imaging Technology (MVIPIT 2023), **Publicity Chair**
- International Conference on Computer and Information Technology for Environmental Sciences (CITES 2023), **Program Chair**
- IEEE International Conference on Computer and Communications (ICCC 2018), **Session Chair**

### 3. Academic organizations Service:

- CCF Multimedia Technology Committee, **Executive Committee Member**
- China Mobile Group Sichuan Company, **Academic and Technical Advisor for the Postdoctoral Research Station**

### 4. Special Issues Service:

- Guest Editor, **Deep Learning Technologies and Their Applications in Image Processing, Computer Vision, and Computational Intelligence**, Sensors (ISSN: 1424-8220) (2025.06.01-2026.03.15)  
Co-organizers: Dr. Honggang Chen (Sichuan University).  
This special issue has received over 10 submissions, 2 papers are accepted for publication, and is currently in progress and continues to welcome new submissions.
- Guest Editor, **Deep Learning Technologies and Their Applications in Image Processing, Computer Vision, and Computational Intelligence**, AI (ISSN: 2673-2688) (2025.06.01-2026.03.15)  
Co-organizers: Dr. Honggang Chen (Sichuan University).  
This special issue has received over 13 submissions, 2 papers are accepted for publication, and is currently in progress and continues to welcome new submissions.
- Guest Editor, **Pattern Recognition and Computer Vision Based on Deep Learning**, Applied Sciences (ISSN: 2076-3417) (2023.02.01-2023.08.20)  
Co-organizers: Prof. Dr. Shuai Li (Shandong University).  
This special issue has received over 4 submissions, 4 papers are accepted for publication. Issue has been published in Vol. 13, 2023.
- Guest Editor, **Deep Learning-Based Computer Vision: Technologies and Applications**, Electronics (ISSN: 2079-9292) (2023.03.01-2023.12.31)  
Co-organizers: Dr. Yuanzhouhan Cao (Beijing Jiaotong University) and Dr. Tao Li (Xihua University).  
This special issue has received over 28 submissions, 10 papers are accepted for publication. Issue has been published in Vol. 12, 2023 and Vol. 13, 2024.

## IEEE MEMBERSHIP

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- IEEE Member information: Member/Customer Number: 93495954

## AREAS OF EXPERTISE AND REPRESENTATIVE PUBLICATIONS

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( \* indicates the corresponding author)

### 1. Image Restoration and Enhancement

- Xin Lin, Yuyan Zhou, Jingtong Yue, **Chao Ren\***, Kelvin C.K. Chan, Lu Qi. Re-boosting self-collaboration parallel prompt gan for unsupervised image restoration[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025, 47(11): 9827-9844. (JCR-Q1, IF=18.6)
- Guanglu Dong, Xiangyu Liao, Mingyang Li, Guihuan Guo, **Chao Ren\***. Channel consistency prior and self-reconstruction strategy based unsupervised image deraining[C]. Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025: 7469-7479.

- Xiangpeng Tian, Xiangyu Liao, Xiao Liu, Meng Li, **Chao Ren\***. Degradation-aware feature perturbation for all-in-One image restoration[C]. Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025: 28165-28175.
- Xiao Liu, Xiaofeng Wang, Shouyi Wang, Haosong Gou, Zhengyong Wang, **Chao Ren\***. GBPG-Net: Global background prior-guided rain and snow image restoration[J]. IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2025, 36(10): 18448-18462. (JCR-Q1, IF=8.9)
- Xin Lin, Jingtong Yue, Sixian Ding, **Chao Ren\***, Qi Lu, Yang Ming-Hsuan. Dual degradation representation for joint deraining and low-light enhancement in the dark[J]. IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2025, 35(3): 2461-2473. (JCR-Q1, IF=11.1)
- Shouyi Wang, Yizhong Pan, Xiaohai He, Zhengyong Wang, **Chao Ren**. PFCPNet: A progressive feature correction and prompt network for robust real-world image denoising[J]. Neurocomputing, 2025: 130221. (JCR-Q1, IF=6.5)
- Lufei Chen, Xiangpeng Tian, Shuhua Xiong, Yinjie Lei, **Chao Ren\***. Unsupervised blind image deblurring based on self-enhancement[C]. Proceedings of the Computer Vision and Pattern Recognition (CVPR), 2024: 25691-25700.
- Xiangyu Liao, Tianheng Zheng, Jiayu Zhong, Pingping Zhang, **Chao Ren\***. Asymmetric Mask Scheme for Self-supervised Real Image Denoising[C]. Proceedings of the European Conference on Computer Vision (ECCV), 2024: 199-215.
- Tingrong Zhang, Xiaohai He, Qizhi Teng, Junxiong Cheng, **Chao Ren**. Spatio-Temporal Adaptive Weighted Fusion Network for Compressed Video Quality Enhancement[J]. IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II), 2024, 71(12): 5064-5068. (JCR-Q1, IF=4.9)
- Xin Lin, **Chao Ren\***, Xiao Liu, Jie Huang, Yinjie Lei. Unsupervised image denoising in real-world scenarios via self-collaboration parallel generative adversarial branches[C]. Proceedings of the International Conference on Computer Vision (ICCV), 2023: 12642-12652.
- Yizhong Pan, Xiao Liu, Xiangyu Liao, Yuanzhouhan Cao, **Chao Ren\***. Random sub-samples generation for self-supervised real image denoising[C]. Proceedings of the International Conference on Computer Vision (ICCV), 2023: 12150-12159.
- Yizhong Pan, **Chao Ren\***, Xiaohong Wu, Jie Huang, Xiaohai He. Real image denoising via guided residual estimation and noise correction[J]. IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2023, 33(4): 1994-2000. (JCR-Q1, IF=11.1)
- Man Qin, **Chao Ren\***, Hong Yang, Xiaohai He, Zhengyong Wang. Blind image denoising via deep unfolding network with degradation information guidance[J]. IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II), 2023, 70(8): 3179-3183. (JCR-Q1, IF=4.9)
- Tingrong Zhang, Qizhi Teng, Xiaohai He, **Chao Ren**, Zhengxin Chen. Multi-Scale Inter-Communication Spatio-Temporal Network for Video Compression Artifacts Reduction[J]. IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II), 70(3): 1229-1233. (JCR-Q1, IF=4.9)
- Jie Huang, Xiao Liu, Yizhong Pan, Xiaohai He, **Chao Ren\***. CasaPuNet: Channel affine self-attention based progressively updated network for real image denoising[J]. IEEE Transactions on Industrial Informatics (TII), 2022, 19(8): 9145-9156. (JCR-Q1, IF=11.7)
- Jie Huang, Zhibo Zhao, **Chao Ren\***, Qizhi Teng, Xiaohai He. A prior-guided deep network for real image denoising and its applications [J]. Knowledge-Based Systems (KBS), 2022, 255. (JCR-Q1, IF=7.6)
- Sen Zhai, **Chao Ren\***, Zhengyong Wang, Xiaohai He, Linbo Qing. An effective deep network using target vector update modules for image restoration[J]. Pattern Recognition (PR), 2022, 122. (JCR-Q1, IF=7.6)
- **Chao Ren\***, Yizhong Pan, Jie Huang. Enhanced latent space blind model for real image denoising via alternative optimization[C]. Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS), 2022: 38386-38399.

- Weiheng Sun, Xiaohai He, **Chao Ren**, Shuhua Xiong, Honggang Chen. A quality enhancement network with coding priors for constant bit rate video coding[J]. Knowledge-Based Systems (KBS), 2022, 258: 110010. (JCR-Q1, IF=7.6)
- **Chao Ren\***, Xiaohai He, Chuncheng Wang, Zhibo Zhao. Adaptive consistency prior based deep network for image denoising[C]. Proceedings of the Computer Vision and Pattern Recognition (CVPR), 2021: 8596-8606.
- **Chao Ren\***, Xiaohai He, Linbo Qing, Yuanyuan Wu, Yifei Pu. Remote sensing image recovery via enhanced residual learning and dual-luminance scheme[J]. Knowledge-Based Systems (KBS), 2021, 222: 107013. (JCR-Q1, IF=7.6)
- **Chao Ren\***, Qizhi Teng, Xiaohai He, Linbo Qing, Truong Q. Nguyen. Compressed image restoration via deep deblocker driven unified framework[J]. Knowledge-Based Systems (KBS), 2021, 228. (JCR-Q1, IF=7.6)
- **Chao Ren\***, Xiaohai He, Linbo Qing, Yuanzhouhan Cao. Deep deblocker driven adaptive iteration scheme for compressed image recovery[C]. Proceedings of the International Conference on Multimedia and Expo (ICME), 2021: 1-6.
- Chuncheng Wang, **Chao Ren**, Xiaohai He, Linbo Qing. Deep recursive network for image denoising with global non-linear smoothness constraint prior[J]. Neurocomputing, 2021, 426: 147-161. (JCR-Q1, IF=6.5)
- Zhengxin Chen, Xiaohai He, **Chao Ren**, Honggang Chen, Tingrong Zhang. Enhanced Separable Convolution Network for Lightweight JPEG Compression Artifacts Reduction[J]. IEEE Signal Processing Letters (SPL), 2021, 28: 1280-1284. (JCR-Q1, IF=3.9)
- Mengdi Sun, Xiaohai He, Shuhua Xiong, **Chao Ren**, Xinglong Li. Reduction of JPEG compression artifacts based on DCT coefficients prediction[J]. Neurocomputing, 2020, 384: 335-345. (JCR-Q1, IF=6.5)
- **Chao Ren\***, Xiaohai He, Truong Q. Nguyen. Adjusted non-local regression and directional smoothness for image restoration[J]. IEEE Transactions on Multimedia (TMM), 2019, 21(3): 731-745. (JCR-Q1, IF=8.4)

## 2. Image Super-Resolution

- Xiao Liu, Xiaofeng Wang, Zhengyong Wang, Xiaohai He, **Chao Ren\***. Real-world blind image super-resolution with mixed and probabilistic scheme based synthetic degradation pipeline[J]. Knowledge-Based Systems (KBS), 2025, 331: 114643. (JCR-Q1, IF=7.6)
- Yingying Zhang, Xiaohai He, Honggang Chen, **Chao Ren**. Depth map super-resolution via learned non-local model and enhanced local regularization[J]. Signal Processing (SP), 2024, 218: 109368. (JCR-Q2, IF=3.6)
- Junxiong Cheng, Shuhua Xiong, Xiaohai He, **Chao Ren**, Tingrong Zhang, Honggang Chen. Nonlocal-guided enhanced interaction spatial-temporal network for compressed video super-resolution[J]. Applied Intelligence (AI), 2023, 53(20): 24407-24421. (JCR-Q2, IF=3.5)
- Honggang Chen, Xiaohai He, Linbo Qing, Yuanyuan Wu, **Chao Ren\***, Ray E. Sheriff, Ce Zhu. Real-world single image super-resolution: A brief review[J]. Information Fusion (IF), 2022, 79: 124-145. (JCR-Q1, IF=15.5)
- Yue Wang, Xiaohong Wu, Xiaohai He, **Chao Ren**, Tingrong Zhang. Deep Feature Fusion Network for Compressed Video Super-Resolution[J]. Neural Processing Letters (NPL), 2022, 54(5): 4427-4441. (JCR-Q3, IF=2.8)
- **Chao Ren\***, Xiaohai He, Yifei Pu, Truong Q. Nguyen. Learning image profile enhancement and denoising statistics priors for single-image super-resolution[J]. IEEE Transactions on Cybernetics (TCYB), 2021, 51(7): 3535-3548. (JCR-Q1, IF=10.5)
- **Chao Ren**, Xiaohai He, Truong Q. Nguyen. Single image super-resolution via adaptive high-dimensional non-local total variation and adaptive geometric feature[J]. IEEE Transactions on Image Processing (TIP),

2017, 26(1): 90-106. (JCR-Q1, IF=13.7)

- **Chao Ren**, Xiaohai He, Qizhi Teng, Yuanyuan Wu, Truong Q. Nguyen. Single image super-resolution using local geometric duality and non-local similarity[J]. IEEE Transactions on Image Processing (TIP), 2016, 25(5): 2168-2183. (JCR-Q1, IF=13.7)
- Honggang Chen, Xiaohai He, **Chao Ren**, Linbo Qing, Qizhi Teng. CISRDCNN: Super-resolution of compressed images using deep convolutional neural networks[J]. Neurocomputing, 2018, 285: 204-219. (JCR-Q1, IF=6.5)
- **Chao Ren**, Xiaohai He, Yifei Pu. Nonlocal Similarity Modeling and Deep CNN Gradient Prior for Super Resolution[J]. IEEE Signal Processing Letters (SPL), 2018, 25(7): 916-920. (JCR-Q1, IF=3.9)

### 3. Image Quality Assessment

- Jingtong Yue, Xin Lin, Zijiu Yang, **Chao Ren\***. Dual-representation interaction driven image quality assessment with restoration assistance[C]. Proceedings of the Winter Conference on Applications of Computer Vision (WACV), 2025.
- Guanglu Dong, Xiangyu Liao, Mingyang Li, Guihuan Guo, **Chao Ren\***. Exploring semantic feature discrimination for perceptual image super-resolution and opinion-unaware no-reference image quality assessment[C]. Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2025: 28176-28187.

### 4. 3D Vision and Generative Models

- Xin Lin, Shi Luo, Xiaojun Shan, Xiaoyu Zhou, **Chao Ren**, Lu Qi, Ming-Hsuan Yang, Nuno Vasconcelos. HQGS: High-quality novel view synthesis with gaussian splatting in degraded scenes[C]. Proceedings of the Thirteenth International Conference on Learning Representations (ICLR), 2025.
- Yuwei Yang, Munawar Hayat, Zhao Jin, **Chao Ren**, Yinjie Lei. Geometry and uncertainty-aware 3d point cloud class-incremental semantic[C]. Proceedings of the Computer Vision and Pattern Recognition (CVPR), 2023: 21759-21768.
- Yuanzhouhan Cao, Yidong Li, Haokui Zhang, **Chao Ren**, Yifan Liu. Learning structure affinity for video depth estimation[C]. Proceedings of the ACM International Conference on Multimedia (ACM MM), 2021: 190-198.
- Junxi Feng, Xiaohai He, Qizhi Teng, **Chao Ren**, Honggang Chen, Yang Li. Reconstruction of porous media from extremely limited information using conditional generative adversarial networks[J]. Physical Review E (PRE), 2019, 100(3): 033308. (JCR-Q1, IF=2.4)

### 5. Multi-modal Learning and Fusion

- Tianheng Zheng, Guanglu Dong, Pingping Zhang, Xiaohai He, **Chao Ren\***. Plug-and-play general image registration for misaligned multi-modal image fusion. IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), 2025, 35(10): 10017-10031. (JCR-Q1, IF=11.1)
- Yanteng Zhang, Qizhi Teng, Xiaohai He, Tong Niu, Lipei Zhang, Yan Liu, **Chao Ren**. Attention-based 3D CNN with Multi-layer Features for Alzheimer's Disease Diagnosis using Brain Images\*[C]. International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), 2023: 1-4.
- Yuanzhouhan Cao, Hui Zhang, Yidong Li, **Chao Ren**, Congyan Lang. CMAN: Learning global structure correlation for monocular 3D object detection[J]. IEEE Transactions on Intelligent Transportation Systems (TITS), 2022, 23(12): 24727-24737. (JCR-Q1, IF=8.4)
- Meiling Wang, Xiaohai He, Luping Liu, Linbo Qing, Honggang Chen, Yan Liu, **Chao Ren**. Medical visual question answering based on question-type reasoning and semantic space constraint[J]. Artificial Intelligence in Medicine (AIM), 2022, 131: 102346. (JCR-Q1, IF=6.2)
- Yixiao Liu, Qizhi Teng, Xiaohai He, **Chao Ren**, Honggang Chen. Multimodal Sensors Image Fusion for Higher Resolution Remote Sensing Pan Sharpening[J]. IEEE Sensors Journal (SJ), 2022, 22(18): 18021-

18034. (JCR-Q1, IF=4.5)

## 6. Efficient Network Architectures

- Xiao Liu, Zhengyong Wang, Hong Yang, Xiaohai He, Haosong Gou, **Chao Ren\***. Transformer-style convolutional network for efficient natural and industrial image superresolution[J]. IEEE Transactions on Industrial Informatics (TII), 2025, 21(7): 5138-5149. (JCR-Q1, IF=10.5)
- Shouyi Wang, Gang Liu, Xiao Liu, Xiangyu Liao, **Chao Ren\***. Efficient image super resolution via Mixed Window and Dimension Interaction[J]. Neurocomputing, 2025, 620: 129211. (JCR-Q1, IF=6.5)
- Zhengxin Chen, Xiaohai He, Tingrong Zhang, Shuhua Xiong, **Chao Ren\***. Dual-stage feedback network for lightweight color image compression artifact reduction[J]. Neural Networks (NN), 2024, 179: 106555. (JCR-Q1, IF=6.3)
- Xiao Liu, Xiuya Shi, Lufei Chen, Linbo Qing, **Chao Ren\***. Efficient parallel multi-scale detail and semantic encoding network for lightweight semantic segmentation[C]. Proceedings of the ACM International Conference on Multimedia (ACM MM), 2023: 2544-2552.
- Xiao Liu, Xiangyu Liao, Xiuya Shi, Linbo Qing, **Chao Ren\***. Efficient information modulation network for image super-resolution[C]. Proceedings of the European Conference on Artificial Intelligence (ECAI), 2023: 1544-1551.

## 7. Image Compression

- Junjun Gao, Qizhi Teng, Xiaohai He, Zhengxin Chen, **Chao Ren**. Mixed entropy model enhanced residual attention network for remote sensing image compression[J]. Neural Processing Letters (NPL), 2023, 55(7): 10117-10129. (JCR-Q3, IF=2.8)
- Zhengxin Chen, Xiaohai He, **Chao Ren**, Tingrong Zhang, Shuhua Xiong. Enhanced Attention Context Model for Learned Image Compression[J]. IEEE Signal Processing Letters (SPL), 2025, 32: 1301-1305. (JCR-Q1, IF=3.9)

## ♡ HONORS AND AWARDS

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- 2025, Sichuan University Annual Outstanding Individual.
- 2025, CVPR Outstanding Reviewer Award.
- 2024, Sichuan University Outstanding Research Talent—Academic Newcomer Award (only 10 recipients annually across all disciplines university-wide), Sichuan University.
- 2024, Third Prize, China Institute of Communications Science and Technology Award (Ranked 3rd), China Institute of Communications.
- 2023, Huawei Spark Award, Huawei Technologies Co., Ltd.
- 2023, Third Prize, Sichuan Provincial Science and Technology Progress Award (Ranked 4th), Sichuan Provincial People's Government, China.
- 2023, Sichuan University Hundred Talents Program, Sichuan University.
- 2023, ICDT Novel Display Technology Competition, National First Prize (Advisor), Society for Information Display (SID) China Chapter, International Conference on Display Technology (ICDT) Organizing Committee, Huawei Device Co., Ltd.
- 2018, 2020, 2024, Sichuan University Sanxin Teaching Award and Wuliangchun Outstanding Faculty Award, 3 times, Sichuan University.
- 2017, National Postdoctoral Innovation Talent Support Program, National Postdoctoral Management Committee, China.
- 2017, Elsevier Outstanding Reviewer Award, 2 times, Elsevier.

## ☞ REVIEWER EXPERIENCE

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- **Reviewers for Journals:** IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Image Processing (TIP), International Journal of Computer Vision (IJCV), IEEE Transactions on Multimedia (TMM), IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), IEEE Transactions on Cybernetics (TCYB), IEEE Transactions on Geoscience and Remote Sensing (TGRS), IEEE Transactions on Artificial Intelligence (TAI), IEEE Transactions on Intelligent Transportation Systems (TITS), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Industrial Informatics (TII), IEEE Transactions on Signal Processing (TSP), IEEE Transactions on Computational Imaging (TCI), IEEE Transactions on Evolutionary Computation (TEC), IEEE Transactions on Cognitive and Developmental Systems (TCDS), IEEE Transactions on Big Data (TBD), IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II), IEEE Transactions on Systems, Man, and Cybernetics: Systems (TSMC), IEEE/CAA Journal of Automatica Sinica (JAS), Optics Express (OE), and Advanced Engineering Informatics (AEI) ...
- **Reviewers for Conferences:** IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE/CVF International Conference on Computer Vision (ICCV), European Conference on Computer Vision (ECCV), Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), International Conference on Machine Learning (ICML), ACM International Conference on Multimedia (ACM MM), Association for the Advancement of Artificial Intelligence (AAAI), Winter Conference on Applications of Computer Vision (WACV) ...